

Chap 14

→ Partial derivatives

exe-1,

find the values of $\frac{df}{dn}$ and $\frac{df}{dy}$ at the point $(4, -5)$ if

$$f(n, y) = n^2 + 3y + y - 1$$
$$= n^2 + 3ny + y - 1$$

so, To find $\frac{df}{dn}$, we treat y as a constant and differentiate with respect to n .

$$\frac{df}{dn} = 2n + 3 \cdot 1 + 0 - 0$$
$$2n + 3 \cdot 1 \cdot y + 0 - 0 = 2n + 3y$$

The value of $\frac{df}{dn}$ at $(4, -5)$ is

$$2(4) + 3(-5)$$
$$8 - 15$$
$$-7$$

$$\frac{df}{dy} = (n^2 + 3y + y - 1)$$

$$\frac{df}{dy} = 0 + 3n \cdot 1 + 1 - 0$$

$$\frac{df}{dy} = 3ny + 1 = 3(4) + 1$$
$$13 \checkmark$$

ex2 Find $\frac{df}{dy}$ as a function of (ny)
 $= y \sin ny$

so

We treat n as a constant and f
as a product of y and $\sin ny$.

73 $f(n, y, z) = n^2 + y^2 - 2z^2 = 0$

$$\Rightarrow \frac{d}{dn} = 2n + \frac{dy}{dy} = 2y + \frac{d}{dz}(-4z) = 0$$

$$\frac{d}{dn} = 2 \quad \frac{d}{dy^2} = 2 + \frac{d}{dz} - 4 = 0$$

74 $f(n, y, z) = 2z^3 - 3(n^2 + y^2)z$

$$\frac{d}{dn} = 2z^3 - 3n^2z - 3y^2z$$

$$\boxed{\frac{d}{dn} = 0 - 6nz - 0}$$

(75)

$$P(ny) = e^{-2y} \cos 2n.$$

$$\frac{d}{dn} = \frac{d}{dn} e^{2y} \cos 2n + \frac{d}{dy} e^{2y} (\cos 2n = 0)$$

$$\frac{d}{dn} e^{-2y} \sin 2n \cdot 2 + \frac{d}{dy} e^{-2y} \cdot (-2) \cos 2n.$$

$$\frac{d}{dn} + 2e^{-2y} \sin 2n + \frac{d}{dy} - 2e^{-2y} \cos 2n.$$

$$\frac{d}{dn} - 4e^{-2y} \cos 2n + \frac{d}{dy} + 4e^{-2y} \sin 2n$$

$$\cancel{-4e^{-2y} \sin 2n} + \cancel{4e^{-2y} \cos 2n}$$

$$0 = 0$$

$$\Rightarrow \frac{1}{(x^2+y^2)^{3/2}} \frac{d}{dx} 2x + \frac{d}{dy} \frac{1}{(x^2+y^2)^{3/2}} = 2y.$$

26, $f(x,y) = \ln \sqrt{x^2+y^2}$
 $\ln (x^2+y^2)^{1/2}$

$f(x,y) = \ln (x^2+y^2)^{1/2}$
 using the log property:

$$\frac{1}{2} \ln (x^2+y^2)$$

Apply chain Rule:

$$f_x = \frac{1}{2} \left(\frac{1}{x^2+y^2} \right) \cdot \frac{d}{dx} x^2+y^2$$

$$\frac{d}{dx} = 2x$$

step 4 substitute the inner derivative back in.

$$f_x = \frac{1}{2} \cdot \frac{1}{x^2+y^2} \cdot 2x$$

$\rightarrow$ simplify 2's.
 $f_x = \frac{x}{x^2+y^2}$

same $\frac{d}{dx}$

we will get

$$f_y = \frac{y}{x^2+y^2}$$

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 $f(x,y) = \tan^{-1} \left(\frac{x}{y} \right)$

→ first partial derivative of f w.r.t $\frac{d}{dx}$.

The derivative of the rule for $\tan^{-1}(u)$ is
 $\frac{d}{du} \left(\tan^{-1}(u) \right) = \frac{1}{1+u^2}$, Here $u = \frac{x}{y}$

$$\frac{\partial}{\partial x} = \frac{1}{1+\left(\frac{x}{y}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{x}{y} \right)$$

Now calculate the inner derivative.

$$\left[\frac{\partial}{\partial x} \left(\frac{x}{y} \right) = \frac{1}{y} \right] \cdot \frac{\partial}{\partial x} (x) = \frac{1}{y} \cdot 1 = \boxed{\frac{1}{y}}$$